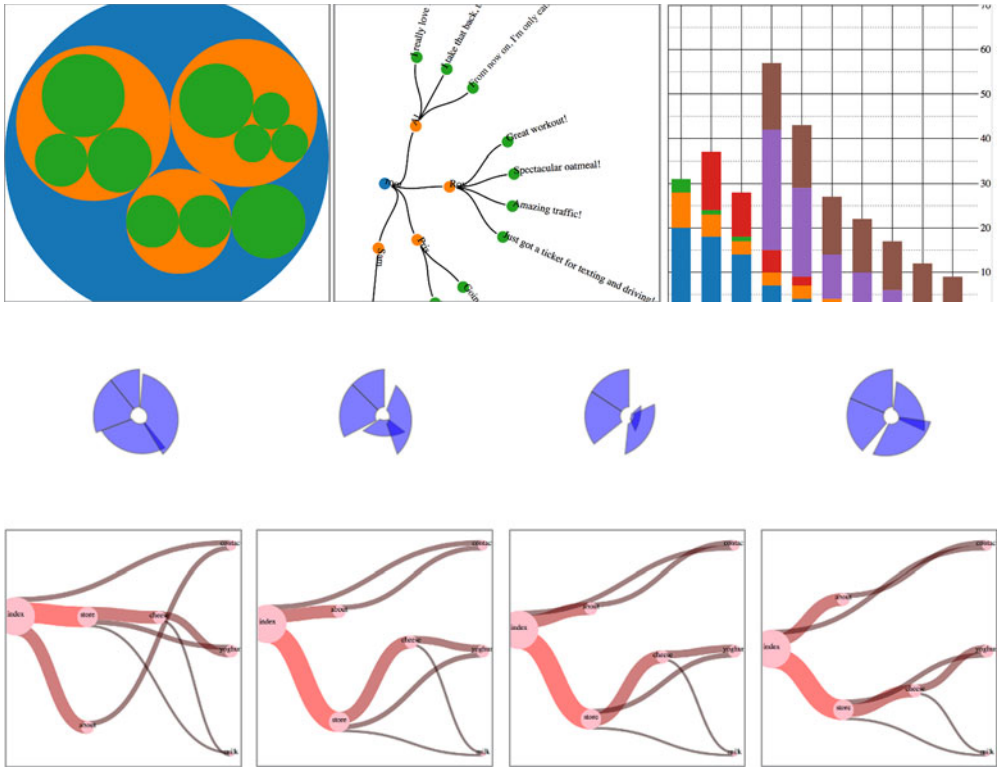


Figure 5.1 Multiple layouts are demonstrated in this chapter, including the circle pack (section 5.3), tree (section 5.4), stack (section 5.5), and Sankey (section 5.6.1), as well as tweening to properly animate shapes like the arcs in pie charts (section 5.2.3).

This chapter gives an overview of layout structure by implementing popular layouts such as the histogram, pie chart, tree, and circle packing. Other layouts such as the chord layout and more exotic ones follow the same principles and should be easy to understand after looking at these. We'll get started with a kind of chart you've already worked with, the bar chart or histogram, which has its own layout that helps abstract the process of building this kind of chart.

5.1 Histograms

Before we get into charts that you'll need layouts for, let's take a look at a chart that we easily made without a layout. In chapter 2 we made a bar chart based on our Twitter data by using `d3.nest()`. But D3 has a layout, `d3.layout.histogram()`, that bins values automatically and provides us with the necessary settings to draw a bar chart based on a scale that we've defined. Many people who get started with D3 think it's a charting library, and that they'll find a function like `d3.layout.histogram` that creates a bar chart in a `<div>` when it's run. But D3 layouts don't result in charts; they result in the settings necessary for charts. You have to put in a bit of extra work for charts, but